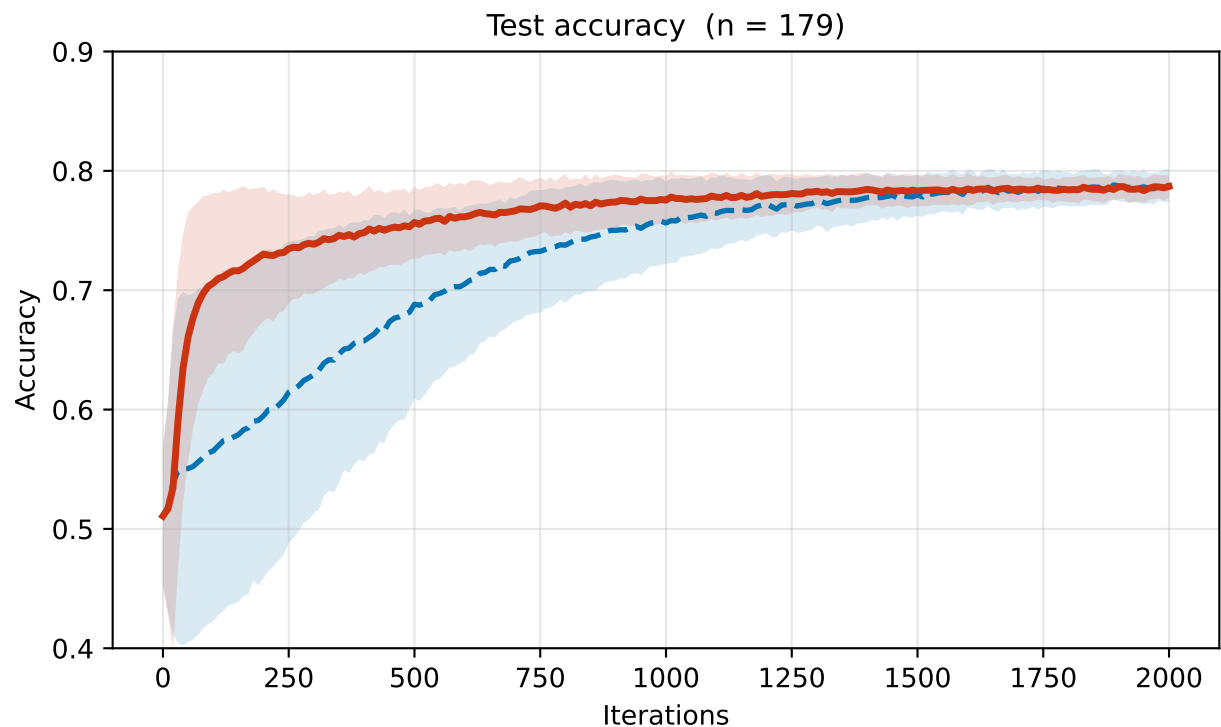
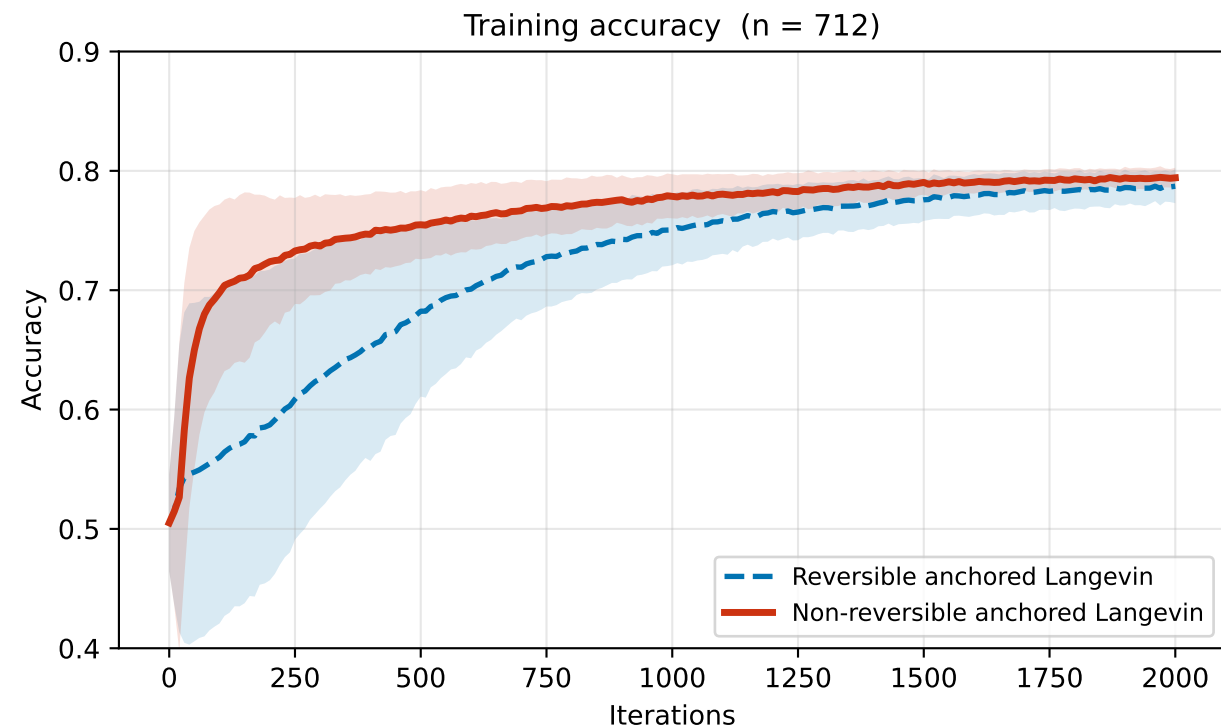


Titanic — non-reversible vs reversible anchored Langevin (ball constraint)



$d = 9$ (intercept + 8 columns: emb_C, emb_Q, sex, pclass, age, sibsp, log_fare, parch); $n_{\text{train}} = 712$, $n_{\text{test}} = 179$; $U = f + g$, $g = 2 \cdot \sum |w_j|$, anchor delta = 0.02; EXACT gradient; $\eta = 1.4 \cdot 10^{-6}$; $s = 12$ (ramped from 0 over the first 10 iterations); $R = 100$.
 Preprocessing: scaling = 'design', order = 'design', signs aligned = True, block reparametrisation (v_{axis} , v_{fast} , v_{slow}) = (1, 1024, 1), (b, e) = (1.0, 0.5). Reference (scikit-learn logistic regression) test accuracy 0.782.
 Lines: across-replicate mean; bands: mean $\pm$ 1 sd of single-iterate accuracy (repeat-run variability). Shared initialisation and noise per replicate. y-axis fixed to (0.4, 0.9), not windowed.